

ZKFly: CUDA-Accelerated Nova for Private Sparse-Graph Execution

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Abstract

ZKFly binds a canonical sparse connectome to a Poseidon root and proves a chain of private weighted CSR evaluations with Nova. The public state is two field elements: a vector commitment and a fixed topology root. The implementation uses HyperKZG, GPU multi-scalar multiplication (MSM), and CUDA arithmetic for Nova’s R1CS operations. Its main optimization vertically concatenates the three matrices (A, B, C) and replaces three same-vector sparse products by one checked device batch. On a V100, a verified 1,024-step, 14,826-constraint fixture falls from 612.609 s to 329.535 s (1.859 $\times$); fusion alone removes 4,092 synchronizations and 4,092 allocations relative to the resident-buffer version. For the 166,700-neuron, 25,582,938-edge artifact, we report exact execution measurements and circuit lower bounds, not an unmeasured full-network proof time.

1 Relation

Let a canonical square CSR graph be

$$G = (N, p, j), \quad 0 = p_0 \leq \dots \leq p_N = M, \quad 0 \leq j_e < N. \quad (1)$$

Columns inside every row are sorted and unique. For private $x \in \mathbb{F}^N, w \in \mathbb{F}^M$, one step constrains

$$y_r = \sum_{e=p_r}^{p_{r+1}-1} w_e x_{j_e}, \quad 0 \leq r < N. \quad (2)$$

The data artifact also carries signed synapse counts c_e and uses

$$w_e = \frac{c_e}{\sum_{k=p_r}^{p_{r+1}-1} |c_k|}, \quad e \in [p_r, p_{r+1}), \quad (3)$$

for its deterministic floating-point execution oracle. In the proof relation, w is an arbitrary private field witness: no training or normalization rule is claimed.

1.1 Canonical roots

Seven little-endian u32 values are packed per BN254 field. With pack_7 , the topology stream is

$$s_G = (L, N, N+1, M) \parallel \text{pack}_7(p) \parallel \text{pack}_7(j), \quad (4)$$

$$L = 4 + \lceil (N+1)/7 \rceil + \lceil M/7 \rceil. \quad (5)$$

For rate $\rho = 11$, $a_0 = 0$, zero-padded chunks $q_i \in \mathbb{F}^{11}$, and domain $D_T = \text{zkfly/poseidon/topology/v1}$,

$$a_{i+1} = \text{Poseidon}_{D_T}(a_i, q_i), \quad R_G = a_{\lceil L/11 \rceil}. \quad (6)$$

The width-13, x^5 Circom parameterization is fixed in host, circuit, and CUDA code. Input and output vectors use the same fold shape under a distinct domain $D_F = \text{zkfly/poseidon/forward/v1}$; write the result as $\text{Com}_F(\cdot)$ [3].

Algorithm 1 Canonical topology commitment

Require: N , CSR arrays $p[0..N]$, $j[0..M]$
Ensure: R_G
 1: assert Equation (1) and sorted-unique columns
 2: $s \leftarrow (L, N, N+1, M) \parallel \text{pack}_7(p) \parallel \text{pack}_7(j)$
 3: $a \leftarrow 0$
 4: **for** $q \leftarrow \text{chunks}_{11}(s)$ **do**
 5: $a \leftarrow \text{Poseidon}_{D_T}(a, q \parallel 0^{11-|q|})$
 6: **return** a

Algorithm 2 Exact-key fused $A/B/C$ sparse product

Require: CSR A, B, C with m rows; packed z ; field (b_F, q_F)
Ensure: Az, Bz, Cz
 1: $K_X \leftarrow (X.p, X.j, X.v, b_F, q_F)$ for $X \in \{A, B, C\}$
 2: $K \leftarrow (K_A, K_B, K_C, m)$
 3: **if** $K \notin \mathcal{C}$ **then**
 4: $\bar{p} \leftarrow A.p \parallel (\text{nnz}(A) + B.p[1:])$
 5: $\parallel (\text{nnz}(A) + \text{nnz}(B) + C.p[1:])$
 6: $\bar{j} \leftarrow A.j \parallel B.j \parallel C.j$; $\bar{v} \leftarrow A.v \parallel B.v \parallel C.v$
 7: $\mathcal{C}[K] \leftarrow \text{Upload}(\bar{p}, \bar{j}, \bar{v}, \text{alloc}(3m))$
 8: $z_D \leftarrow \text{Upload}(\text{Montgomery}(z))$
 9: $o_D \leftarrow \text{SpMV}(\mathcal{C}[K], z_D)$ ▷ one launch
 10: $o \leftarrow \text{DownloadAndSync}(o_D)$ ▷ one boundary
 11: **return** $(o[0:m], o[m:2m], o[2m:3m])$

1.2 Recursive state

For step t , Nova carries

$$z_t = (u_t, R_G), \quad u_t = \text{Com}_F(x_t), \quad z_{t+1} = (\text{Com}_F(y_t), R_G). \quad (7)$$

The witness is (x_t, w_t) and the circuit enforces

$$\mathcal{R}_G(z_t, z_{t+1}; x_t, w_t) = \begin{cases} u_t = \text{Com}_F(x_t), \\ y_t = W_t x_t \text{ under Equation (2)}, \\ u_{t+1} = \text{Com}_F(y_t), \quad R'_G = R_G. \end{cases} \quad (8)$$

Consecutive witnesses satisfy $\text{Com}_F(x_{t+1}) = \text{Com}_F(y_t)$. Nova folds the resulting relaxed R1CS instances [5]. The primary commitment backend is HyperKZG, derived from the Gemini/KZG polynomial-commitment line [2, 4]; MSM is dispatched to Blitzar.

2 CUDA arithmetic boundary

For R1CS matrices $A, B, C \in \mathbb{F}^{m \times d}$ and packed $z \in \mathbb{F}^d$, Nova repeatedly requests

$$(a, b, c) = (Az, Bz, Cz). \quad (9)$$

The compatibility interface has a three-product default; the CUDA override constructs

$$\bar{M} = \begin{bmatrix} A \\ B \\ C \end{bmatrix} \in \mathbb{F}^{3m \times d}, \quad \bar{M}z = \begin{bmatrix} Az \\ Bz \\ Cz \end{bmatrix}. \quad (10)$$

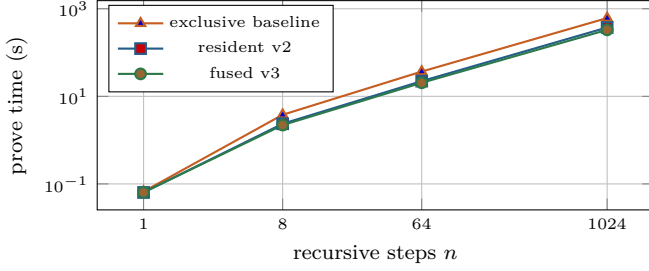


Figure 1: HyperKZG + GPU-MSM proving time; all outputs verified.

Table 1: Verified fixed-circuit profile ($C = 14,826$, $V = 14,825$).

n	exclusive (s)	resident (s)	fused (s)	v3 verify (s)
1	0.067	0.064	0.064	0.301
8	3.793	2.363	2.206	0.126
64	36.814	22.241	20.388	0.127
1024	612.609	373.514	329.535	0.127

Setup: baseline 1.124 s, resident 1.149 s, fused 1.187 s. At $n = 1024$: fused/exclusive = 0.5381 (1.859 $\times$); fused/resident = 0.8823 (1.133 $\times$).

Cache identity. No digest key is used:

$$K_M = (p_M, j_M, v_M, b_F, q_F), \quad K = (K_A, K_B, K_C, m). \quad (11)$$

The bounded FIFO retains at most 8 fused CSR batches; reusable arithmetic outputs are keyed by exact field-element length.

Equivalence. For $0 \leq r < m$, the first, second, and third row intervals of $\bar{M}$ are exactly rows r of A, B, C , while all column indices remain unchanged. Therefore Equation (10) holds entrywise. Equality in Equation (11) implies byte-identical canonical CSR input, field width, and modulus; hence a cache hit selects the same device matrix. A regression test checks offset shifting and ordering; every timed recursive proof is verified against its public root and endpoint commitments.

3 Operation model

Let $n \geq 1$ be recursive steps and $k = n - 1$. Each non-base Nova fold performs two R1CS evaluations. Thus

$$\begin{aligned} N_{\text{SpMV}}^{\text{logical}} &= 6k, & N_{\text{SpMV}}^{\text{batch}} &= 2k, \\ N_{\text{lin}} &= 4k, & N_{\text{cross}} &= 2k. \end{aligned} \quad (12)$$

Resident, unfused execution has

$$S_{v2} = B_{v2} = 12k, \quad A_{v2} = 22k, \quad (13)$$

while fused execution has

$$S_{v3} = B_{v3} = 8k, \quad A_{v3} = 18k. \quad (14)$$

Here S, B, A are synchronizations, physical arithmetic batches, and device buffer allocations. At $n = 1024$, fusion removes $4k = 4,092$ synchronizations and allocations.

4 Measurements

Platform. Tesla V100-SXM3-32GB (SM 7.0), CUDA 12.6.85, driver 580.159.03. Compiler: `rustc 1.96.0-nightly`; `cuda-oxide 0.2.0`. Profiling uses a single serialized CUDA context and reusable Nova parameters. Setup is excluded from prove/verify columns.

Table 2: Fused $n = 1024$ prover telemetry.

operation	logical calls	physical batches	time (s)
$A/B/C$ SpMV	6,138	2,046	77.657
vector linear combination	4,092	4,092	86.097
cross term	2,046	2,046	67.736
MSM	2,047	2,047	36.268
MSM scalars		30,347,798	
CSR cache		6,138 hit / 0 miss	
output cache		6,138 hit / 0 miss	
allocations / synchronizations		18,414 / 8,184	

Table 3: Full-graph numeric forward pass; this is not proving time.

CUDA executor	kernel (ms)	end-to-end (ms)	Gedge/s
resident row/thread	13.575	13.722	1.864
edge products + row reduce	4.630	4.770	5.363

Both: 100 measured iterations after 10 warmups, device-resident topology/weights/input/output, maximum absolute error = 0 against the deterministic f32 CPU oracle. The edge-parallel run reports 100.158 GiB/s effective traffic.

4.1 Full artifact: execution, then capacity

The source connectome reports 166,700 neurons [1]. The derived deterministic artifact has

$$\begin{aligned} N &= 166,700, & M &= 25,582,938, \\ |\text{CSR}| &= 205,330,308 \text{ bytes.} \end{aligned} \quad (15)$$

and topology root

$$413\text{eb9f07080ce45} \dots 46\text{caaa9a01b0f323}. \quad (16)$$

The implemented estimator uses a conservative lower bound $C_H = 2,174$ constraints per width-13 Poseidon permutation. Therefore

$$\begin{aligned} L_T &= 4 + \lceil (N+1)/7 \rceil + \lceil M/7 \rceil = 3,678,525, \\ S_T &= \lceil L_T/11 \rceil = 334,412, \\ C_T &= S_T C_H = 727,011,688, \end{aligned} \quad (17)$$

$$\begin{aligned} Q &= \lceil N/11 \rceil = 15,155, & P_F &= 2Q = 30,310, \\ C_F &= P_F C_H + M + N = 91,643,578, \end{aligned} \quad (18)$$

$$\begin{aligned} W_F &= M + 2N = 25,916,338 \text{ fields} \\ &= 829,322,816 \text{ bytes (canonical)}. \end{aligned} \quad (19)$$

One topology registration plus T forward ticks has

$$\begin{aligned} C_{\text{total}}(T) &\geq C_T + T C_F, \\ C_{\text{total}}(1000) &= 92,370,589,688. \end{aligned} \quad (20)$$

5 Result boundary

proved: $z_0 \xrightarrow[\mathcal{R}_G^n]{\text{Nova}} z_n$;
measured: $n \leq 1024$, $C = 14,826$, HyperKZG + GPU-MSM + CUDA;
estimated: C_T, C_F, W_F for Equation (15);
not claimed: full-artifact or distributed proof, training, weight commitment, or 4.770 ms proof.

Nova’s recursive chain is sequential in the current implementation. Graph tiles, Poseidon chunks, and MSMs admit parallel sub-work, but no distributed proof aggregation protocol is implemented or measured. A full forward proof also requires a larger commitment key, a memory study, and an end-to-end run; the small-circuit timing in Table 1 is not extrapolated.

Table 4: Measured vs. projected proof scope.

quantity	measured fixture	full artifact
neurons / edges	3 / 4	166,700 / 25,582,938
primary constraints	14,826	$\geq 91,643,578$
constraint ratio	$1\times$	$\geq 6,181.27\times$
Poseidon vector chunks	1	15,155
Powers-of-Tau scale	2^{15} available	$\geq 2^{27}$ projected
recursive proof	verified	not run
wall-clock proof time	measured	not claimed

Receipt integrity. The fused standard and smoke receipts have SHA-256 prefixes/suffixes `72859e53...7e6c6716` and `0930b641...c30b26`, respectively. The profile schema records logical SpMV calls separately from physical batches; all standard cases and the 12-neuron, two-commitment-chunk regression verified.

References

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